

Homework-2

Packages

Question 1

```
install.packages("artsy")
```

The difference between `install.packages` and `library` is that `install.packages` is plugged into your console on R and permanently downloads a package onto your computer, it only needs to be done once as well. Unlike `library` which is needed whenever you want to call specific data set values without listing the data set. This is needed whenever a new file/project is being used.

Question 2

```
library(aRtsy)
```

Collatz conjecture is the theory that by doing a constant equation to every single number ever, it will eventually arrive at one. There are two parts of it, if the number is even, $n/2$ until you achieve 1. But if the number is odd, the equation is $3n + 1$ until the fraction is even and then do $/2$ again. Continue this process until you arrive at 1.

Question 3

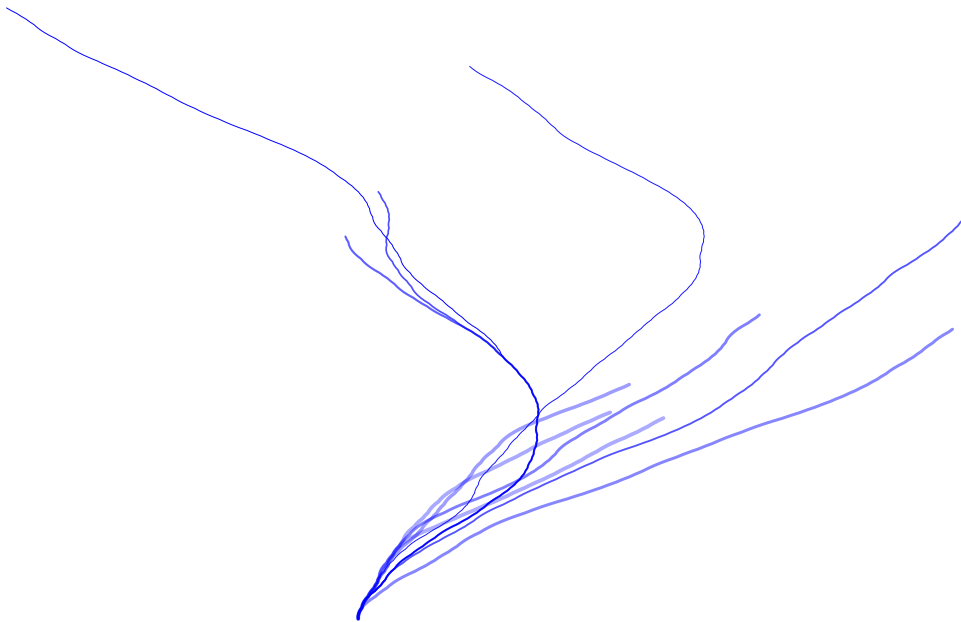
a.

This function is taking 10 random positive integers, IE the $n=10$ and then does the two collatz equations to them until they get to one and depending on what the equation does to the number, it makes the line move in different directions. The color “blue” also made the lines blue.

b.

1. Because it gives different random positive integers each time you run the code
2. Because each time the code is run, it generates a different photo
3. Set.seed can be used to ensure the graph looks the same because it designates a specific random number generation which applies to every running of the code. So it doesn't change.

```
canvas_collatz(  
  set.seed(14),  
  colors = c("blue"),  
  n = 10  
)
```



Question 4

This code appears to do the same thing as the previous function except it makes the function affect 1 through 20 positive random integers and then stays with that amount for every running of the code. This causes the lines to not change no matter how many times the code is run. Set.seed wouldn't be necessary here because the lines don't need to be set to a specific formation, they already are.

```
canvas_collatz(  
  colors = c("blue"),  
  n = 1:20  
)
```

